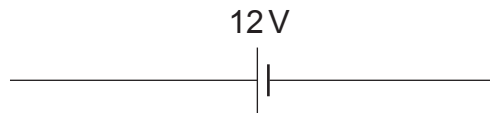


4. Jonathon is investigating how the resistance of a thermistor varies with temperature.

(a) Complete the circuit diagram used to measure the resistance of a thermistor.

[3]



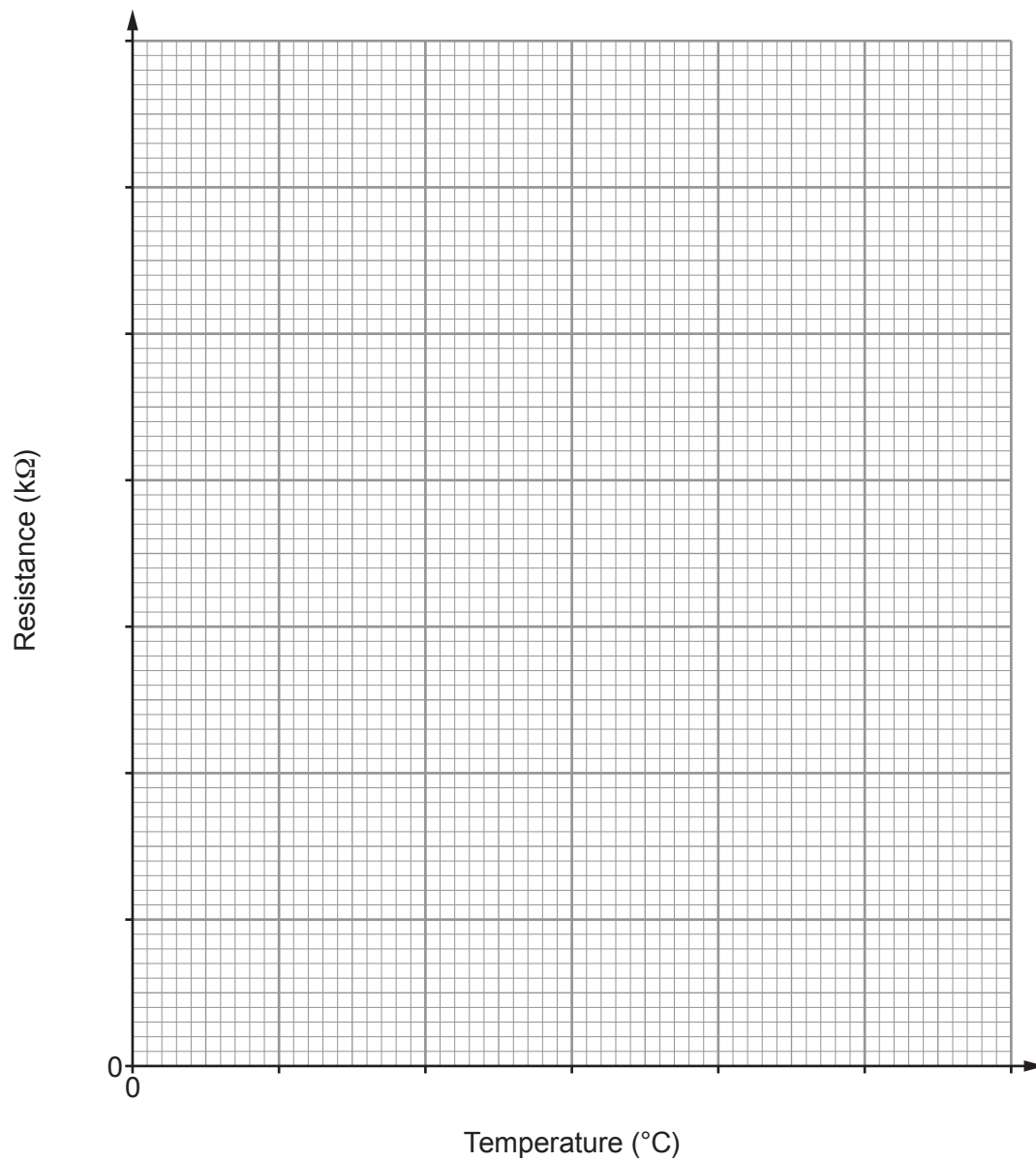
(b) Jonathon's results are shown in the table below.

Temperature (°C)	Current (mA)	Voltage (V)	Resistance (kΩ)
20	1.01	12.0	11.9
40	2.21	12.0	5.4
60	4.78	12.0	2.5
80	8.51	12.0	1.4
100	19.95	12.0	0.6



- (i) Plot the data on the grid and draw a suitable line.

[3]



- (ii) Describe the relationship shown by the graph.

[2]

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(iii) Use the graph and a suitable equation from the list below to answer each of the following questions.

resistance (Ω) = $\frac{\text{voltage (V)}}{\text{current (A)}}$

power = voltage $\times$ current

power = current² $\times$ resistance

I. Calculate the current through the thermistor at 30 °C. [4]

current =mA

II. Calculate the power developed in the thermistor at 30 °C. [2]

power =W

(c) Thermistors are found in refrigerators to control the temperature.

Describe how the experiment needs to be modified to test if this thermistor would be suitable for a refrigerator. [2]

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